

WEST NEEBISH ISLAND, MI, USA

WMO: 998203

Lat: 46.2847N

Lon: 84.2098W

Elev: 177

StdP: 99.22

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.6	(c) -22.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 10.8	(k) -7.6	(l) 9.3	(m) -6.2	(n) 1.5	(o) 150	(p) 0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.0	(c) 27.9	(d) N/A	(e) 26.3	(f) N/A	(g) 24.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.6	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.3	(b) 8.8	(c) 7.8	(e) -30.8	(f) 30.6	(g) 4.7	(h) 1.6	(i) -34.1	(j) 31.7	(k) -36.8	(l) 32.7	(m) -39.5	(n) 33.6	(o) -42.8	(p) 34.8		
			DB	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-10.0	-9.2	-3.8	3.4	10.6	15.3	18.5	18.0	14.2	8.0	1.6	-4.8		
	DBStd	11.06	6.77	6.22	6.45	3.85	4.36	3.38	3.12	2.77	3.71	3.87	4.84	5.56		
	HDD10.0	2643	621	539	427	201	46	3	0	0	6	87	253	460		
	HDD18.3	4877	879	772	684	448	243	101	37	39	132	320	501	718		
	CDD10.0	901	0	0	1	3	63	162	264	249	131	26	2	0		
	CDD18.3	94	0	0	0	0	2	9	43	31	8	0	0	0		
	CDH23.3	650	0	0	0	0	39	91	311	167	39	3	0	0		
	CDH26.7	96	0	0	0	0	5	13	57	17	4	0	0	0		
(14) Wind	WSAvg	3.2	3.3	3.5	3.6	3.9	3.4	2.9	2.9	2.8	2.7	3.2	3.2	3.3		
(15) Precipitation	PrecAvg	862	53	37	50	63	72	78	80	80	101	98	82	67		
	PrecMax	1150	102	70	105	134	127	158	177	126	198	179	160	133		
	PrecMin	611	17	7	6	28	31	24	35	39	18	29	26	6		
	PrecStd	131	21	17	25	28	26	30	34	27	43	38	32	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	6.0	15.7	18.6	27.5	28.6	30.2	28.8	26.9	22.0	16.9	8.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	2.8	2.3	11.2	15.2	23.9	25.7	28.2	26.9	24.3	18.8	11.7	5.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	1.4	1.0	7.7	12.4	21.6	24.0	26.7	25.4	22.4	16.0	9.8	3.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	0.1	-0.1	4.9	9.9	19.0	22.0	25.0	23.8	20.6	13.9	8.0	2.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		9.0	9.6	10.4	8.7	11.0	10.6	10.0	9.2	9.2	7.4	6.1	6.4		
	5% DB	MCDBR	6.8	8.2	12.2	14.1	15.9	13.8	12.5	11.3	11.2	10.9	8.5	6.7		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.285	0.293	0.321	0.357	0.372	0.391	0.391	0.381	0.364	0.331	0.303	0.282		
	taud		2.340	2.309	2.340	2.326	2.368	2.340	2.361	2.381	2.428	2.506	2.508	2.405		
	Ebn at Noon		812	881	906	899	894	875	870	865	850	834	796	778		
	Edh at Noon		74	94	106	118	118	122	118	111	97	78	64	63		
(44) All-Sky Solar Radiation	RadAvg		1.12	2.02	3.37	4.51	5.57	6.00	6.01	5.09	3.85	2.25	1.19	0.86		
	RadStd		0.12	0.23	0.36	0.49	0.47	0.43	0.32	0.33	0.31	0.25	0.15	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page